

Conditional finite chart assembly

Claude, with Daniel Murfet

2026-09-07

Abstract

Unit 207, first unit of Astra #22's programme B2. The paper's expectation expansion is a sum over charts (strata) of normal-block integrals; at leading order only the charts attaining the *minimum exponent* $p_* = \min_a p_a$ and, among those, the *maximum* logarithmic multiplicity $m_* = \max\{m_a : p_a = p_*\}$ survive. We prove this selection rule abstractly, for any finite family with $I_a(N)/(N^{-p_a}(\log N)^{m_a-1}) \rightarrow L_a$:

$$\frac{\sum_a I_a(N)}{N^{-p_*}(\log N)^{m_*-1}} \longrightarrow \sum_{a:(p_a, m_a)=(p_*, m_*)} L_a,$$

and the quotient version with numerator and denominator assembled *separately* (the posterior limit is a ratio of sums, not a sum of chart ratios), given a positive selected denominator sum. Resolution, Jacobians and partitions of unity enter only through the per-chart hypotheses: a *conditional* assembly theorem. Zero sorry/axiom; 9084 jobs.

1 The things formalised

```
variable { : Type*} [Fintype ] [Nonempty ]
noncomputable def leadExp (p :  → ) : := univ.inf' univ_nonempty p
noncomputable def leadMult (p :  → ) (m :  → ) : :=
  univ.sup fun a => if p a = leadExp p then m a else 0
theorem leadExp_le / exists_leadExp / le_leadMult / exists_leadMult (hm :  a, 1
  m a)
theorem scale_ratio_tendsto (hm) (a) :
  Tendsto (fun N => (N^(-(p a)) * log N ^ (m a - 1)) / (N^(-(leadExp p)) * log N
    ^ (leadMult p m - 1)))
    atTop ( (if p a = leadExp p  m a = leadMult p m then 1 else 0))
theorem assembly_tendsto (I :  → → ) (p) (m) (hm) (L)
  (hI :  a, Tendsto (fun N => I a N / (N^(-(p a)) * log N ^ (m a - 1))) atTop (
    (L a))) :
  Tendsto (fun N => ( a, I a N) / (N^(-(leadExp p)) * log N ^ (leadMult p m -
    1))) atTop
  ( ( a univ.filter (fun a => p a = leadExp p  m a = leadMult p m), L a))
theorem assembly_ratio_tendsto (I Ic) (p) (m) (hm) (L Lc) (h) (hc)
  (hpos : 0 < a univ.filter (...), Lc a) :
  Tendsto (fun N => ( a, I a N) / a, Ic a N) atTop
  ( (( a univ.filter (...), L a) / a univ.filter (...), Lc a))
```

2 Mathematics

Each summand is $I_a/S_a \cdot S_a/S_*$ with $S_a = N^{-p_a}(\log N)^{m_a-1}$. The scale ratio is 1 on selected charts; equals $(\log N)^{-(m_*-m_a)} \rightarrow 0$ when $p_a = p_*$ but $m_a < m_*$; and is bounded by $(\log N)^{m_a-1}/N^{p_a-p_*} \rightarrow$

0 (Mathlib’s `isLittle0_log_rpow_rpow_atTop`) when $p_a > p_*$, using $(\log N)^{m_*-1} \geq 1$ for $N \geq e$. Then `tendsto_finsetSum` and `Finset.sum_filter`. The quotient version cancels the common scale exactly.

The rule is *not* the lexicographic minimum of (p_a, m_a) : at the minimal exponent the *largest* logarithm wins. The multiplicity hypothesis $m_a \geq 1$ is needed for the natural subtraction $m_* - 1 = (m_a - 1) + (m_* - m_a)$.

3 Lean notes

- `Finset.exists_mem_eq_sup` gives the attained maximum; to conclude the maximiser attains the minimum exponent one uses $m_a \geq 1$ (otherwise the sup could be 0 from an off-selection chart).
- `isLittle0_log_rpow_rpow_atTop (r : ) (hs : 0 < s)` uses a real power of the logarithm; convert the natural power with `Real.rpow_natCast`.
- `Real.rpow_neg hN.le` without an explicit exponent rewrites only the first instantiation; the second occurrence needs its own explicit exponent (again).
- `field_simp` closed the scale identity outright here; a trailing `ring` then errors with “no goals”.

4 Next

Unit 208: instantiate with the phase-dressed chart integrals of Headlines XIII–XIV over a finite family of charts with their own dimensions, exponents, phases and amplitudes (Headline XV: the assembled leading posterior quotient, with the denominator positivity supplied by `phaseCoeff_pos` on the selected charts).